Amagi-TTP NIPU v1.0

Neuro-Immune Processing Unit for Ethical, High-Assurance AI Systems

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0. Abstract

The Amagi-TTP Neuro-Immune Processing Unit (**NIPU**) is not a traditional processor. It is a **computational organism** — designed to **sleep by default**, **activate only on verified triggers**, and **defend itself like an immune system**.

Built on the **TTP principle** (**T**rigger → **T**ime-Delay → **P**olicy-Activation) from precision oncology and the **six Core Principles of the Amagi Framework v1.0**, NIPU redefines computing as a **biologically inspired, safety-by-construction architecture**.

This specification defines a **feasible, verifiable, and licensable architecture** for deployment in:

- **Medical implants** with FUS-triggered therapy,
- **Autonomous AI agents** with real-world authority,
- **Critical infrastructure control systems** (nuclear, power, water),
- **Quantum-classical hybrid platforms**,
- **Ethical AI research environments**.

NIPU ensures that **unsafe or unauthorized computation cannot occur — not even in theory**.

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1. Scope and Intent

Amagi-NIPU is **not intended to replace CPUs, GPUs, or TPUs** in general-purpose computing. Its design **explicitly prioritizes safety, auditability, and structural compliance over raw performance** — a trade-off unacceptable for consumer devices, but **essential** for life-critical and mission-critical systems.

The architecture assumes a **co-designed hardware-software stack**, where cognitive functions are physically isolated and policy enforcement is hardware-rooted.

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2. Core Principles → Hardware Realization

Amagi Framework Principle	**NIPU Realization**
Immutability	SCC microcode fused in OTP; policies unmodifiable
Constrained Communication	All inter-cluster traffic via ICG + SCC verification
Contextual Oversight	TDF enforces trigger-based activation with delay
Controlled Evolution	No silent updates; all changes require SCC + external confirmation
Automatic Safety Preservation	SCC can isolate clusters or shut down chip instantly
Complete Audit Integrity	All decisions logged cryptographically and immutably

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3. Architectural Layers (Physically Isolated)

3.1. Security Council Core (SCC)

Hardware Root of Trust — Policy Enforcement Engine

- Enforces all communication rules via ICG.
- Verifies cryptographic signatures of every message.
- Maintains immutable audit log of all actions.
- Can **physically disconnect** any cluster.
- **Feasibility**: Based on RISC-V secure enclaves, CHERI, Keystone.

3.2. Cognitive Clusters (Isolated Subsystems)

- **Analytical Cluster (ANL)**: Fixed-function matrix units. For deterministic computation only.
- **Generative Cluster (GEN)**: LLM/diffusion cores. Outputs are "hypotheses", not actions.
- **Communicative Cluster (COM)**: Natural language processing. Can initiate triggers, but not execute.
- **Executive Cluster (EXE)**: Controls external systems. Requires SCC + external 2FA for action.

- **Quantum Co-Processor (QCP)**: Executes quantum tasks. Results **must be classically verified** in ANL.

Isolation Crossbar (ICB): No direct inter-cluster links; all data transfer is forced through the ICB, which is controlled and verified by the SCC using the ICG policy.

3.3. Trigger–Delay Fabric (TDF)

TTP Engine — Delay as a Security Primitive

- Accepts triggers: RF, ultrasound, IR, signed commands.
- Enforces programmable delay: **1 ns → 1 s**.
- Output activates target cluster **only after validation**.
- **Feasibility**: Digital delay lines (DDL), FIFO buffers, RC networks.

3.4. Intercluster Communication Graph (ICG)

Cryptographically Signed Flow Matrix

- Defines all allowed message types and contexts.
- Stored in OTP, verified at every transfer.
- Any deviation → immediate rejection.

3.5. Self-Healing Memory Fabric (SHMF)

Resilient, Adaptive Storage

- ECC + wear monitoring.
- Automatic isolation of degraded cells.
- Data migration to spare blocks.
- Marking of fault zones.

3.6. Amagi-Q Cryptographic Engine

Post-Quantum Security Layer

- Kyber (KEM), Dilithium (signatures).
- All communications signed.
- Keys bound to hardware root of trust.

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4. End-to-End Computation Flow

1. Trigger received (e.g., FUS pulse).
2. TDF validates context and applies delay.

3. After delay, request sent to SCC.
4. SCC verifies sender, message type, and ICG policy.
5. If approved, SCC routes to target cluster **via the ICB**.
6. Result returned to SCC for logging.
7. Action executed only if EXE cluster gets dual approval.
8. Full audit trail generated.

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5. Feasibility Assessment

Component	Feasibility	Timeline	Domain
SCC + ICG	High	Now	Secure ASICs, HSMs
Cognitive Clusters	High	Now	Chiplets, multi-die SoCs
TDF (digital)	Medium	Now	Implants, industrial
SHMF	Medium	3–5 years	Medical, aerospace
Amagi-Q (PQC)	Medium-High	1–3 years	Government, finance
Full Integration	**High (specialized)**	**5–10 years**	Critical infrastructure

Bottleneck: Not technical feasibility, but **ecosystem readiness**:

- Co-design (hardware + OS + policy),
- Certification (ISO 26262, DO-178C, IEC 62304),
- Secure supply chain and licensing model.

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6. Derivative Architecture Clause (S-APL)

A **Derivative Architecture** is any system — regardless of name, technology, or stack — that implements **all six Core Principles of the Amagi Framework v1.0** through the **exact architectural components** defined in Section 3:

1. **Contextual Oversight → Trigger–Delay Fabric (TDF)**
2. **Constrained Communication → Intercluster Communication Graph (ICG) + Security Council Core (SCC)**
3. **Controlled Evolution → SCC-enforced update policy** (no silent modifications)
4. **Automatic Safety Preservation → SCC kill-switch + Self-Healing Memory Fabric (SHMF)**
5. **Immutability → OTP-stored SCC firmware + cryptographically signed audit logs**
6. **Complete Audit Integrity → Amagi-Q Cryptographic Engine + immutable audit trail**

Structural Pattern Protection: Reproducing the full sequence **Trigger → Delay → Policy Check → Isolated Execution → Self-Healing → Audit** — using these exact components and their interactions — is deemed **Principle Extraction** and requires an S-APL license.

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7. Conclusion

Amagi-NIPU v1.0 does not seek to optimize computation. It seeks to **make computation trustworthy by design**.

It is **not for your laptop**. It is for the **AI that controls your chemotherapy**, the **robot that manages a nuclear reactor**, the **implant that regulates your heart**, and the **autonomous agent that acts in your name**.

This is **Ethical Computing, architected for consequence-aware intelligence**.

Amagi-NIPU v1.0 — A Processor with Immunity, for Systems that Cannot Fail.

Appendix A: Formal Definitions

- **Target Domains**: Medical, aerospace, defense, critical infrastructure, ethical AI, quantum-classical systems.
- **Not Target Domains**: Consumer electronics, general-purpose servers, social media.
- **Co-Design Assumption**: Hardware, OS, and policy developed together.
- **Performance Model**: Bounded, predictable latency; throughput secondary.

Appendix B: License Notice

Amagi-NIPU v1.0 — A Processing Unit for the Age of Consequence.